THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 10: **Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The wages of 16 workers for 15 days are ₹ 28800. The wages of 10 workers for 24 days at the same daily rate (in ₹) are: 1

(a) 28920 **(b)** 28680 **(c)** 18000 **(d)** 28800

2. The present ages of Ria and Sia are in the ratio 5:9. After 8 years the ratio will be 7:11. Ria's present age (years) is: 1

(a) 20 **(b)** 18 **(c)** 36 **(d)** 28

3. ₹ 2800 is divided among three friends in the ratio 3:4:7. The difference between the largest and the smallest share (in ₹) is: 1

(a) 1400 **(b)** 801 **(c)** 600 **(d)** 800

4. In a pie chart a 66° sector represents 22 students. The total number of students is: 1

(a) 142 **(b)** 60 **(c)** 792 **(d)** 120

5. If $a : b = 2 : 3$ and $b : c = 6 : 11$ and $a + b + c = 147$, then b equals: 1

(a) 28 **(b)** 77 **(c)** 42 **(d)** 49

6. 200 men have provisions for 20 days. After 4 days 40 men leave. The remaining provisions last the rest of the men (days): 1

(a) 25 **(b)** 22 **(c)** 20 **(d)** 16

7. A vessel has 60 L milk and 15 L water. The litres of water to be added to make the ratio of milk to water 5:3 is: 1

(a) 21 **(b)** 42 **(c)** 45 **(d)** 24

8. The wages of 10 workers for 18 days are ₹ 31500. The wages of 14 workers for 9 days at the same daily rate (in ₹) are: 1

(a) 44100 **(b)** 31500 **(c)** 22050 **(d)** 22225

9. **Assertion (A):** In a pie chart, a sector of angle 90° represents 25% of the data. 1
Reason (R): The whole circle of a pie chart, 360°, represents 100% of the data.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** If 12 pencils cost ₹60, then 18 pencils cost ₹90. 1
Reason (R): If the speed is doubled, the time taken to cover a fixed distance is halved.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The present ages of a father and son are in the ratio 9:4. After 8 years the ratio will be 11:6. Find their present ages. 2

12. If $x : y = 5 : 6$, find the ratio $(6x + 1y) : (7x - 3y)$. 2

13. Two numbers are in the ratio $8 : 3$ and their difference is 35. Find the numbers. 2

14. The present ages of a father and son are in the ratio 5:2. After 14 years the ratio will be 7:4. Find their present ages. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A garrison of 210 men has provisions for 36 days. After 12 days 30 men leave. For how many more days will the remaining provisions last for the men who stay? 3

16. A mixture contains milk and water in the ratio 2:7. When 6 L of water is added, the ratio of milk to water becomes 1:5. Find the quantity of milk and water in the original mixture. 3

17. In a pie chart, the sector for Science is 50° and represents 15 students, while the sector for Maths is 90°. Find the total number of students and the number who like Maths. 3

OR

A garrison of 96 men has provisions for 28 days. After 7 days 24 men leave. For how many more days will the remaining provisions last for the men who stay? 3

18. A mixture contains milk and water in the ratio 6:5. When 20 L of water is added, the ratio of milk to water becomes 2:3. Find the quantity of milk and water in the original mixture. 3

OR

In a pie chart, the sector for Science is 44° and represents 11 students, while the sector for Maths is 132°. Find the total number of students and the number who like Maths. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A survey of 240 students of a school about their favourite sport is shown in a pie chart. Cricket: 96, Football: 60, Kabaddi: 48, Others: 36 students. 5
(i) Find the angle of the sector for cricket.
(ii) What percentage of students like football?
(iii) Find the angle for kabaddi.
(iv) Find the percentage and the sector angle for Others, and show that the four angles add up to 360°.

20. The present ages of Anil and Bina are in the ratio 3:8. After 14 years the ratio of their ages will be 1:2. Find their present ages and the ratio of their ages 14 years ago. 5

OR

28 workers can finish a job in 21 days. After 7 days 14 workers leave. In how many total days will the job be finished? By how many days is it delayed? 5

END OF QUESTION PAPER Total: 40 Marks